

Saw so far: vector space, subspace, span, SCV is called lin indep if no-nontrivial linear combination of elts of S is 0

V is called a f.dim v.space if V has a finite spanning set

basis of V is a lin indep set that spans V

$\dim V = |\text{any basis}| \rightsquigarrow$ Still have to show this is well defined

Thm ① Any f.dim v.space V has a basis

② Any two bases of V have the same cardinality

- We proved both for $\mathbb{R}^n$: $\{v_1, \dots, v_k\} \subset \mathbb{R}^n$ is a basis for $\mathbb{R}^n$
 $\Leftrightarrow$ $\left\{ \begin{array}{l} \text{RREF(matrix with cols } v_1, \dots, v_k) \text{ has a pivot in} \\ \text{each row } (\Leftrightarrow \text{spanning}) \text{ \& each col } (\Leftrightarrow \text{lin indep}) \end{array} \right. \therefore \dim \mathbb{R}^n = n$
- As a result $\left\{ \begin{array}{l} \text{any set of } n \text{ lin indep vectors in } \mathbb{R}^n \text{ is a basis.} \\ \text{————//—— vectors in } \mathbb{R}^n \text{ that span } \mathbb{R}^n \text{ ———//——} \end{array} \right.$

For general f.d.v.sp V , strategy to build a basis

① Keep adding lin indep vectors till they span

Claim Any lin indep set I can be extended to give a basis

Pf. Take finite spanning set $S = \{u_1, \dots, u_k\}$.

Let $I_0 = I$

[Yesterday I was confused and made this harder than it needs to be!]

Define I_1 as follows

If $u_1 \notin \text{Span}(I)$ then "add u_1 to I " i.e. $I_1 = I_0 \cup \{u_1\}$
or "keep I same" i.e. $I_1 = I_0$

Keep going. Note that each I_j is lin indep and

$\text{span}(I_k)$ contains S hence $\text{span}(I_k) = V$. Hence I_k is a basis for V .

③ Take a spanning set S and keep deleting unnecessary vectors until remainder is lin indep.

Thus we have proved ① for any fdim v-space (in two ways)

proof of ②: we will give 2 proofs

For the first proof we'll use the notion of **isomorphism**

Imp note: A basis $B := \{v_1, \dots, v_n\}$ of a v-space V , defines a "coordinate" system (address system) on V .

How? Any $u \in V$ can be written **uniquely** as a lin combination of $v_1, \dots, v_n$.

- Because $\{v_1, \dots, v_n\}$ spans V $\exists c_i$ s.t. $u = \sum_{i=1}^n c_i v_i$
- Why are c_i unique?

$$\text{If } u = \sum c_i v_i = \sum d_i v_i$$

$$\text{then } 0 = \sum c_i v_i - \sum d_i v_i = \sum (c_i - d_i) v_i$$

$\therefore$ By lin indep of $\{v_1, \dots, v_n\}$ each $c_i = d_i$

We often impose an order on B and then $\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ is

the coordinate vector (in $\mathbb{R}^n$) of u wrt the ordered basis B .

Def Suppose V and W are vector spaces.

A function $f: V \rightarrow W$ is called an isomorphism if

- 0) f is a bijection ("dictionary")

$$1) f(v_1 + v_2) = f(v_1) + f(v_2) \quad 2) f(\lambda v) = \lambda f(v)$$

Rmk If f is an isomorphism so is f^{-1}

$$\left. \begin{aligned} f^{-1}(w_1 + w_2) &\stackrel{?}{=} f^{-1}(w_1) + f^{-1}(w_2) \\ f^{-1}(f(v_1) + f(v_2)) &= \text{LHS} \quad \left| \begin{array}{c} v_1 + v_2 \\ \text{RHS} \end{array} \right. \end{aligned} \right\} \begin{array}{l} w_1 = f(v_1) \text{ for unique } v_1 \\ w_2 = f(v_2) \text{ --- " --- } v_2 \end{array}$$
$$= f^{-1}(f(v_1 + v_2)) = v_1 + v_2$$

If there is an isomorphism from V to W , we call V & W isomorphic vector spaces.

Rmk If $f: \mathbb{R}^c \rightarrow \mathbb{R}^r$ given by $f(x) = Ax$ for $r \times c$ matrix
then f satisfies ① & ②. So f is an isom
 $\Leftrightarrow \text{RREF}(A) = I$ (so $r=c$) $\Leftrightarrow A$ is invertible

If $V \xrightarrow{f} W$ is an isom then any sentence in V
(that involves only the vspace structure of V) stays valid
when transported to W by means of f and vice versa
e.g.

- u is a lin comb of $S \subset V \Leftrightarrow f(u)$ is a lin comb of $f(S)$
- U subspace of $V \Leftrightarrow f(U)$ is a subspace of $W = f(V)$
- S is lin indep in $V \Leftrightarrow f(S)$ is lin indep in W
- S is a basis of $V \Leftrightarrow f(S)$ is a basis of W

Back to pf of ② for f.v.s.p V

Let $\{v_1, \dots, v_n\}$ be a basis $\leadsto$ We have an isomorphism

$$\mathbb{R}^n \xrightarrow{f} V$$

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \mapsto \sum c_i v_i$$

If $\{u_1, u_2, \dots, u_k\}$ is also a basis of V , then get, using ^{another}

$$\mathbb{R}^k \begin{matrix} \xleftarrow{g} \\ \xrightarrow{g'} \end{matrix} V \begin{matrix} \xrightarrow{f} \\ \xleftarrow{f'} \end{matrix} \mathbb{R}^n$$

Observe/ If X, Y, Z are V -spaces s.t.
Exercise $\left\{ \begin{array}{l} X \xrightarrow{f} Y \text{ is an isom} \\ Y \xrightarrow{g} Z \text{ is an isom} \end{array} \right\}$, then $X \xrightarrow{g \circ f} Z$ is also an isomorphism.

Why does this complete the proof of ②?

Added after class

The more abstract proof of ② is still left to do.

Do read and try to follow the sketch written in blue at the end of yesterday's notes.